Gonzalo Díaz Amor

Senior Data Scientist — LLM Evaluation & Statistical/Causal ML · Healthcare & Pharma

Palencia, Spain (CET) · Remote · Open to EOR or contractor

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Professional Summary

Data Scientist with 6 years across pharma, healthtech, and mobility, specialized in LLM evaluation (LLM-as-a-Judge, RAG) and statistical/causal ML (Double ML, Causal Forest, survival analysis). Production deployments on AWS and GCP with FastAPI, Docker, and Airflow, plus clinical programming (CDISC) for EMA submissions.

Relevant Experience

AI/ML Research Engineer

BeWay (rejoined by invitation) | *Jan 2026 - Present* | Remote

- **LLM-as-a-Judge for decision audits:** designed a judge that audits narrative–quantitative consistency in multi-criteria decision analyses (BOCR/AHP), mitigating position bias with per-replica shuffling and 5-replica majority voting.
- Reached 91.7% unanimous verdicts across replicas over 1,272 comparisons, showing the judge was stable enough to trust at scale.
- Flagged 19% of comparisons as narrative–quantitative contradictions and routed only those to targeted human review, replacing exhaustive manual checks.
- Benchmarked 36 runs across 2 models (GPT-4o, GPT-5.2) and 3 temperatures, refining the prompt over 5 human-reviewed iterations.
- **RAG evidence builder (PoC, lead developer):** end-to-end pipeline with human-in-the-loop source curation: multi-format ingestion, LLM-as-a-Judge source scoring, hybrid retrieval (dense + BM25, RRF), and multi-agent synthesis with mandatory chunk-level citations (LangChain orchestration, Gemini API, local Phi-3 fallback).
- Pre-registered the evaluation protocol (confirmatory vs. exploratory hypotheses, power analysis, bootstrap CIs, Benjamini-Hochberg): LLM–expert source selection met the criterion (F1 = 0.762 vs. 0.70 threshold) after I caught a caching bug that produced a spurious F1 = 0.45. Generalized these statistics into the open-source package llm-judge-stats.
- Reported failed hypotheses without post-hoc threshold changes: LLM–human quality scores correlated weakly (Spearman rho = 0.19, 95% CI -0.09 to 0.45); chunk-level citations cut the audited hallucination rate from 57.5% to 14.0%, still above the pre-registered 5%.
- Shipped a React/TypeScript + FastAPI web app with live SSE progress for non-technical analysts, Dockerized, with pytest and CI (mypy, flake8).

Tech stack: Python, R, OpenAI API, Gemini API, Hugging Face transformers, ChromaDB, FastAPI, React, TypeScript, AWS (S3), Docker, GitHub Actions.

AI/ML Specialist Data Scientist

Daiichi-Sankyo (Contract) | *Oct 2025 - Dec 2025* | Remote

- Led the discovery phase of a Causal AI engagement linking German environmental data to ~100,000 patients, screening hundreds of candidate drivers of patient outcomes with Double ML and Causal Forest.
- Designed the causal analysis framework (confounder selection, estimation strategy) that set the methodology for the subsequent effect-estimation phase.
- Built the data infrastructure and high-performance analytics scripts for Google Cloud deployment.

Tech stack: Python, Google Cloud.

Behavioral Data Scientist

BeWay | *Jun 2025 - Oct 2025* | Remote

- Led Geo Data Science analysis across hundreds of features to measure product penetration over 60M+ tickets and inform strategic decisions.

- Owned survival and funnel analytics on 10k+ customers to identify conversion barriers and accelerate time-to-first-purchase.

Tech stack: R, Python, Quarto, AWS (S3, Athena, Redshift), PostgreSQL, Git.

Data Science Specialist

Minsait | *Nov 2023 - Jun 2025* | Remote

- Built an enterprise OMOP-CDM data lake integrating clinical and operational sources from 3 major Madrid hospitals.
- Owned RESTful ML/AI inference APIs with FastAPI, enabling embeddings, vector search, and LLM-powered analytics for downstream applications.
- Deployed 100+ NLP/ML models and tools in production with Docker and Airflow, reducing manual handoffs.

Tech stack: R, Python, Docker, FastAPI, PowerShell, GitLab, MySQL, PostgreSQL, Spark, Airflow, OHDSI.

Data Scientist

TFS HealthScience | *Jun 2022 - May 2023* | Remote

- Served as programming lead for a Phase IV study, validating SDTM and ADaM workflows.
- Delivered propensity-score-matched comparative analyses on real-world evidence (RWE) from 20,000 patients over 7 years, avoiding a new clinical trial, plus regulatory-ready statistical plans and study reports.

Tech stack: R, SAS, Python, CDISC, OHDSI.

Data Scientist

Animal Data Analytics (formerly PigCHAMP Pro Europa) | *Sep 2020 - Jun 2022* | Remote

Segovia-based provider of herd-management software, benchmarking, and data science services for the swine and animal health industry, serving international clients.

- Delivered analytics and ML solutions for 5+ international clients across clinical and commercial engagements, working remotely in English.
- Led a 2-person team to build a regression model that reduced RMSE by 7% vs. the baseline model, with fully reproducible reporting in RMarkdown.
- Co-authored peer-reviewed research as second author ("Feed intake patterns of modern genetics lactating sows", *Porcine Health Management*, 2023), owning data curation and analysis: applied unsupervised ML (PAM clustering, silhouette-based model selection in R) to electronic-feeder data from 585 sows and 1,058 lactations, identifying 5 feeding patterns.
- Quantified the effect of those feeding patterns on reproductive outcomes with GLMMs and non-parametric tests in SAS.

Tech stack: R, Python, SAS, RMarkdown, PyTorch, TensorFlow, MongoDB.

Selected Projects

llm-judge-stats | *Open-source Python package* | github.com/gondiam/llm-judge-stats

- Statistics to check whether an LLM judge can be trusted: Cohen's/Fleiss' kappa, ICC, F1/MCC with bootstrap CIs, sample-size planning, and position-bias shuffling; tests validated against published values.
- Applied to the public MT-Bench human judgments (1,814 items): GPT-4 vs. human-majority kappa = 0.51 (95% CI 0.47-0.54), on par with the human-human kappa of 0.50, with 15.8% position-inconsistent verdicts. The same data gives kappa = 0.38 under a different label order.

sft-dpo-eval | *Pre-registered post-training study* | github.com/gondiam/sft-dpo-eval

- Fine-tuned Qwen3-1.7B on PubMedQA with SFT followed by DPO (LoRA, TRL) on a free Kaggle T4 GPU, committing the analysis plan before any training run.
- SFT raised accuracy from 38.6% to 71.7% (+33.1 pts, 95% CI 27.4–38.9). DPO added no gain (–1.0 pt) and was reported as a null result. An exploratory analysis traced a ~7-pt MMLU drop mainly to repetition loops.
- Three seeds, paired comparisons with 95% CIs and Holm correction, run with llm-judge-stats.

Education

- **MSc. Big Data & Data Science** | VIU (2025) · Thesis: Trading strategy optimization with Deep Learning & quantitative analysis
- **BSc. Statistics** | UVa – Valladolid, Spain (2020)

Certifications

- **AI Agents Intensive Course** — Google x Kaggle *(2025)*
- **6.86x: Machine Learning with Python – From Linear Models to Deep Learning** — MITx *(2021)*
- *More certifications on LinkedIn.*

Technical Skills

- **LLM Evaluation & Statistics:** LLM-as-a-Judge, bias mitigation, pre-registered evaluation protocols, power analysis, bootstrap CIs, inter-rater agreement (Cohen's kappa, ICC), Spearman, F1/MCC, Benjamini-Hochberg.
- **GenAI & LLMs:** RAG pipelines, hybrid retrieval (dense + BM25, RRF), multi-agent pipelines, embeddings, vector search (ChromaDB), prompt engineering, LangChain, LLM-based architectures, LLM fine-tuning (SFT, DPO, LoRA, TRL).
- **Languages & ML:** Python, R, SQL, SAS, PyTorch, TensorFlow, Scikit-Learn, Pandas, NumPy, Causal ML (Causal Forest, Double ML), survival analysis, predictive modeling.
- **MLOps & Cloud:** Docker, Apache Airflow, AWS, GCP, Git (GitHub, GitLab), CI/CD.
- **Data Engineering & Databases:** ETL pipelines, MySQL, PostgreSQL, MongoDB.
- **Clinical Data Standards:** CDISC (SDTM, ADaM), OMOP-CDM (OHDSI).
- **AI-Assisted Development:** Claude Code, Cursor, Google Antigravity, Warp.
- **Languages:** Spanish (Native), English (professional; working language for 5+ years in international projects).